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Juniper Automated Test Station

A Python/Flask application driving the Juniper test bench: the **Vitrek V71 Hi-Pot Tester** over USB or RS-232, a **Siglent SDL1020X-E** DC electronic load, and an always-on **Siemens LOGO! PLC** thermal rig. Results are stored in SQLite and exported to a branded Excel workbook for SharePoint. Built at Juniper Design.

> 📄 **Print-ready PDF:** `docs/pdf/README.pdf`

Features

- **Two access levels** — operators run approved sequences with no login; admins define them. See `docs/access-control.md`
- **Saved test sequences** — reviewed, versioned definitions rather than parameters typed at the bench
- **Instrument verification (PVD)** — daily performance verification against a Vitrek APVD, with the actual measured values recorded, not just a pass light. See `docs/pvd-verification.md`
- **USB and RS-232** — USB via the Silicon Labs CP2110 HID-to-UART DLL; RS-232 via pyserial
- **Full V7X command set** — ACW, DCW, IR, GB, CONT, PAUSE, HOLD steps
- **DC load + thermal rig** — SDL1020X-E load batteries, PEC-0063 thermal qualification, continuous 1 Hz sensor recording
- **Web UI** — nothing to install on operator machines, just open the page
- **Excel export** — colour-coded workbook covering test sessions, thermal qualification and instrument verification

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Quick Start

```
pip install -r requirements.txt
python app.py
```

Open `http://localhost:5000`.

Connect the V71: set `CONFIG MENU → INTERFACE = USB` on the front panel and plug in the USB-B cable, or `= RS232` with a fully-wired 9-wire null-modem cable (hardware RTS/CTS handshaking is required).

First run

1. Click **Set up admin** in the top bar and choose a password. It is stored only as a PBKDF2 hash — keep the password itself in 1Password.
2. As admin, open **HiPot** → **Sequence Builder** and create at least one sequence. Operators cannot run anything until one exists.
3. Baseline the PVD profile before relying on verification verdicts — see `docs/pvd-verification.md`.

Instruments

Instrument	Interface	Notes
Vitrek V71 HiPot	USB HID-to-UART, or RS-232	ACW, DCW and CONT modes only — no IR, no GB
Siglent SDL1020X-E DC Load	TCP/LAN, VISA or USB CDC	
Siemens LOGO! PLC + thermal rig	Modbus TCP	Always on, independent of instrument selection

Instrument selection is mutually exclusive — one at a time — but the thermal rig runs in parallel with any instrument test, and sensor data is continuously recorded so results can be correlated against the timeline.

Project Structure

```
├── app.py                Flask app: REST API + embedded UI for every page
├── auth.py               Roles, admin password hashing, @admin_required
├── v71_driver.py         V71 USB/serial driver (ctypes + pyserial)
├── sdl1020x_driver.py    Siglent DC load driver
├── pvd_test.py           Instrument verification runner
├── pvd_profiles.json     Verification profiles (APVD-74 / APVD-7X)
├── test_battery.py       Multi-step DC load test batteries
├── pec0063_test.py       PEC-0063 thermal qualification
├── database.py           SQLite schema and CRUD
├── excel_export.py       Branded Excel workbook generator
├── plc/                  LOGO! PLC driver, thermal controller, rig_config.json
├── hipot_results.db      SQLite database (auto-created; gitignored)
└── docs/
    ├── access-control.md
    ├── pvd-verification.md
    ├── wiring-guide.md
    ├── hmi-architecture.md
    ├── plc-ladder-logic.md
    ├── breadboard-prototype.md
    ├── sourcing-list.md
    └── V7x_Series_Operating_Manual.pdf
```

REST API

Routes marked **admin** require an admin session; everything else is available to operators. Full detail in the two docs above.

Auth and setup

Method	Endpoint		Description
GET	`/api/auth/status`		Current role
POST	`/api/auth/login`		Elevate to admin
POST	`/api/auth/logout`		Drop to operator
POST	`/api/auth/set_password`		Set or change the admin password
GET/POST	`/api/settings/connection`	admin to write	Saved connection settings

Test sequences

Method	Endpoint		Description
GET	`/api/sequences`		List saved sequences
POST	`/api/sequences`	admin	Create
PUT	`/api/sequences/<id>`	admin	Update (bumps revision)
DELETE	`/api/sequences/<id>`	admin	Retire (`?hard=1` deletes)

HiPot

Method	Endpoint		Description
POST	`/api/connect`		Connect (operators use the saved settings)
POST	`/api/disconnect`		Disconnect
POST	`/api/hipot/run_sequence`		Run a saved sequence
POST	`/api/hipot/run`	admin	Run ad-hoc steps
POST	`/api/hipot/abort` · `/api/hipot/cont`		Abort / continue
GET	`/api/hipot/status` · `/api/hipot/live`		Run state and live V/A/Ω

Verification, results and export

Method	Endpoint		Description
POST	`/api/pvd/start` · `/api/pvd/ack` · `/api/pvd/stop`	admin	Run a verification
POST	`/api/pvd/baseline/promote`	admin	Adopt baseline measurements as nominals
GET	`/api/pvd/results` · `/api/pvd/verification_status`		Verification history and currency
GET	`/api/sessions` · `/api/session/<id>`		Test history
GET	`/api/export` · `/api/export/<id>`		Excel export

USB Communication Details

The V71 uses a **Silicon Labs CP2110 HID-to-UART bridge**:

- `SLABHIDtoUART.dll` + `SLABHIDDevice.dll` (x64, from `software/drivers/`)
- USB VID `4292` (0x10C4), PID `34869` (0x8835)

- 115200 baud, 8N1, RTS/CTS flow control
- ASCII commands terminated with `\r\n`

Key commands (Section 6 of the operating manual):

Command	Description
`*IDN?` / `*RST` / `*ERR?`	Identify / reset / read error register
`NOSEQ`	Clear and activate sequence #0
`ADD,ACW,…` `ADD,DCW,…` `ADD,IR,…` `ADD,GB,…` `ADD,CONT,…`	Add a test step
`ADD,PAUSE,…` `ADD,HOLD,…`	Timed pause / operator prompt
`RUN` · `ABORT` · `CONT` · `RUN?`	Execution control
`RSLT?` · `STAT?` · `STEPRSLT?,<n>`	Overall, per-step and detailed results
`MEASRSLT?,<AMPS|VOLTS|OHMS>`	Live measurement

There is **no remote command for AUTO PVD, SELF TEST or CAL VERIFY** — those are front-panel only, which is why verification is implemented as a programmed sequence. See `docs/pvd-verification.md`.

Excel Export

One workbook, several sheets:

- **Summary** — all test sessions, sortable and filterable
- **Per-session** — full step results, colour-coded
- **PEC-0063 Thermal Results** — thermal qualification against the UL limits
- **PVD Verification** — instrument verification, one row per measured point

Suitable for direct upload or auto-sync to SharePoint.

Packaging & deployment

The bench machine (**HI-POT-TEST**) does not run this from source — it runs a packaged single-file executable, deployed and kept current by the Juniper inventory server.

- **Build the exe:** `powershell -ExecutionPolicy Bypass -File build-exe.ps1 -Version <x.y.z>`. This produces `dist/HiPotController-<x.y.z>.exe` (PyInstaller **onefile**, windowed) and prints its SHA256 + size. The build bundles `static/`, `pvd_profiles.json`, `plc/rig_config.json`, and the x64 Silicon Labs CP2110 DLLs at the path `v71_driver.py` loads them from. `build/` and `dist/` are gitignored — the exe is a build artefact, published to the server, not committed.
- **Persistent data (important for packaging).** Under a onefile exe, `os.path.dirname(__file__)` is the temporary extraction dir that Windows deletes on exit, so the app writes its **results DB**

and the **mutable** **`pvd_profiles.json`** (baseline promotion rewrites it) to `C:\ProgramData\Juniper\HiPotController` instead — see `paths.py` (`data_dir()` / `bundled_dir()`). Read-only bundled resources (`rig_config.json`, brand assets) are read from the bundle. Running from source is unchanged: `data_dir()` resolves to the repo, so dev keeps its files in place.
- **Deploying to the fleet:** the exe is uploaded to the inventory server and the catalog package **"Juniper HiPot Controller"** (device-scoped to HI-POT-TEST) is bumped to the new version, with its install script pinned to the new exe's hash. The agent installs to `C:\Program Files\Juniper\HiPot Controller` on next check-in.

Notes

- The DLLs must be present at `software/drivers/USB_DLLs_and_Headers/USB DLLs and Headers/x64/`, and the app must run as **x64** to match them
- After a DUT breakdown on USB the V71 may disconnect and reconnect; the driver surfaces the error — just reconnect from the UI
- RS-232 requires hardware handshaking; a 3-wire cable will not work
- `hipot_results.db` is gitignored: it holds the admin password verifier and the session signing key, neither of which belongs in the repository